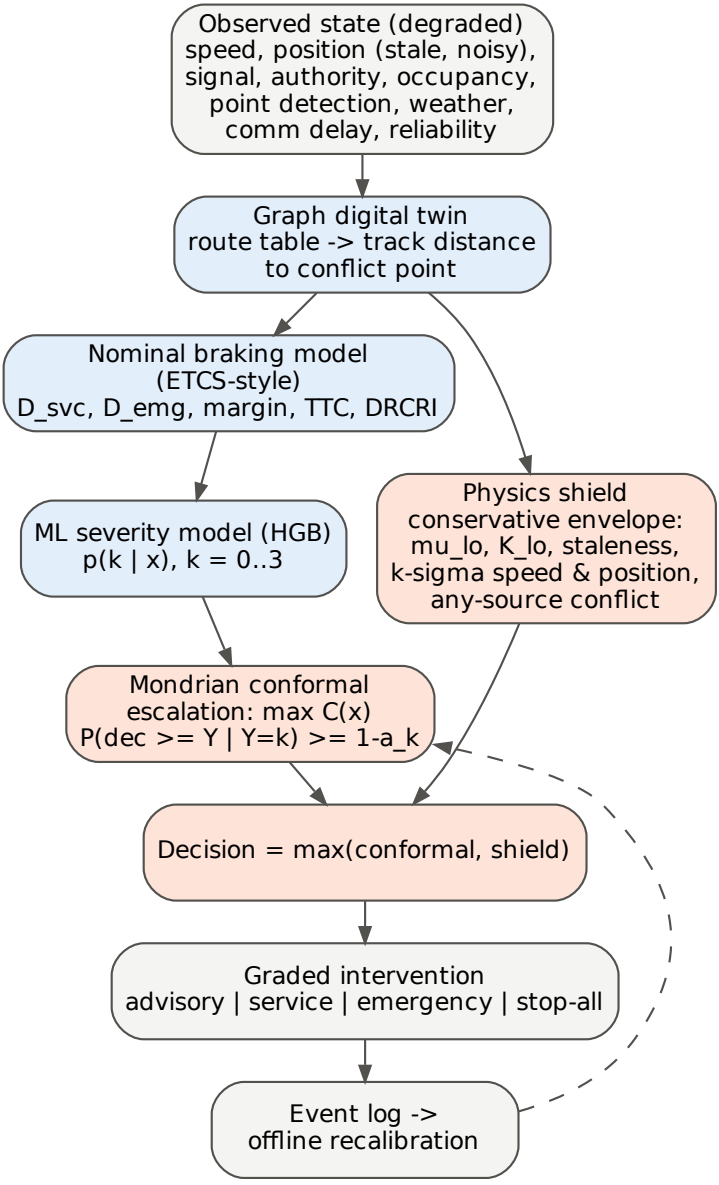
Observed state (degraded)
speed, position (stale, noisy),
signal, authority, occupancy,
point detection, weather,
comm delay, reliability
Graph digital twin
route table -> track distance
to conflict point
Nominal braking model
(ETCS-style)
D_svc, D_emg, margin, TTC, DRCRI
ML severity model (HGB)
p(k | x), k = 0..3
Physics shield
conservative envelope:
mu_lo, K_lo, staleness,
k-sigma speed & position,
any-source conflict
Mondrian conformal
escalation: max C(x)
P(dec >= Y | Y=k) >= 1-a_k
Decision = max(conformal, shield)
Graded intervention
advisory | service | emergency | stop-all
Event log ->
offline recalibration